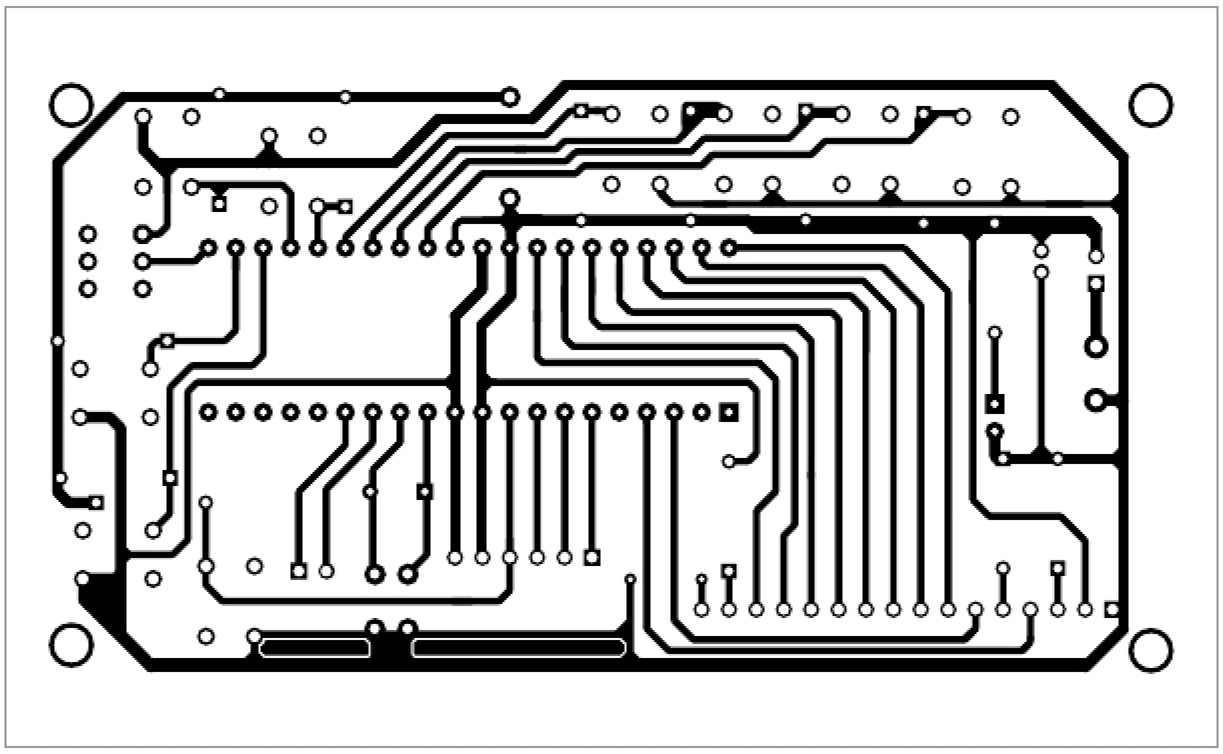

Fig. 2: Actual-size PCB layout of the serial data debugger

PARTS LIST	
Semiconductors:	
IC1	- ATmega16A MCU
LCD1	- 16X4 LCD
LED1	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1, R2, R4, R13	- 1-kilo-ohm
R3	- 10-kilo-ohm
R5-R12	- 4.7-kilo-ohm
Capacitors:	
C1	- 10µF, 16V electrolytic
C1-C4	- 0.1µF ceramic disk
Miscellaneous:	
CON1	- 2-pin terminal connector
CON2	- 6-pin connector
CON3	- 2-pin connector
S1	- On/off switch
S2, S4-S11	- Tactile switch
	- 40-pin IC base
	- 5V power bank of 2000mAh

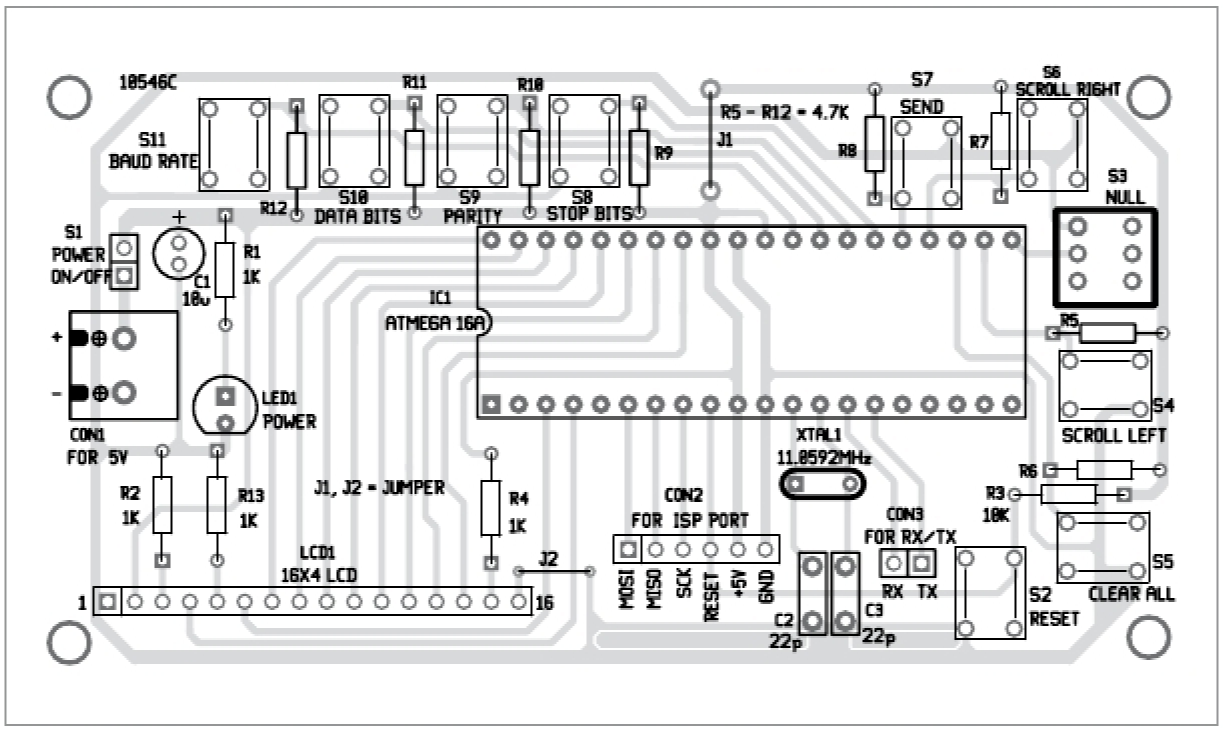

Fig. 3: Components layout for the PCB

To scroll through data, use scroll left (S4) and scroll right (S6) buttons. Data in the buffer can be studied comfortably. By pressing clear all (S5) button, all data in the buffer and screen will get cleared, and new data will be received and displayed on LCD1. The number of characters received is displayed on the right-bottom of LCD1.

To send a data stream, press send (S7) button. String/data stored in send_data (in the code) will be transmitted through Tx pin of the MCU.

Setup and self-test

After successfully assembling the circuit on the breadboard or PCB, write SerialDataDebugger.hex file to ATmega16A (IC1) and set fuse bits for crystal option (ckopt = 1, sut0 = 0,sut1 = 1,cksel0 = 1,cksel1 = 0,cksel2 = 0,cksel3 = 0) through ISP port using AVR programmer. (Refer Atmel data sheet for full details.) Note that a fresh MCU has a default value of fuse bytes, which is equal to 0X99E1 in hexadecimal.

EFY Note
The source code of this project is available for free download at *source.efymag.com*

For testing the programmed MCUs, connect Rx and Tx pins of IC1 (MCU1) to Tx and Rx pins of MCU2, respectively. Power on both units, and press send (S7) button on MCU2. Data/string stored in send_data buffer should appear on LCD1 of MCU1. Disconnect Rx and Tx pins, and the serial data debugger will be ready to use.

For power supply, use a 5V power bank of 2000mAH or higher capacity, which is rechargeable and makes the complete circuit handy.

Construction

An actual-size PCB layout of the serial data debugger is shown in Fig. 2 and its components layout in Fig. 3. Assemble the circuit on the designed PCB and connect 5V DC across CON1. CON2 is used for programming Atmega16A using any ISP programmer. CON3 is used to connect transmitter (Tx) and receiver (Rx) pins from another MCU.

After assembling and testing the circuit, enclose it in a suitable cabinet. Fix the switches, LCD1 and LED1 in front of cabinet. Keep 5V, 2000mAh power bank inside the cabinet.

EFY note. Rx of IC1 should be connected to Tx of another MCU and vice-versa. EFY

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